MIL-M-38510/5C
20 MAY 1985
SUPERSEDING
MIL-M-0038510/5B(19)
1 December 1975 and
MIL-M-38510/5A
6 December 1971
(See 6.7)

MILITARY SPECIFICATION

MICROCIRCUITS, DIGITAL, TTL,

AND-OR-INVERT GATES, MONOLITHIC SILICON

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the detail requirements for monolithic, silicon, TTL, AND-OR-INVERT logic gating microcircuits. Two product assurance classes and a choice of case outlines and lead finishes are provided and are reflected in the complete part number.

1.2 Part number. The complete part number shall be in accordance with MIL-M-38510.

1.2.1 Device type. The device type shall identify the circuit function as follows:

Device type	Circuit
01	Expandable Dual 2-wide, 2-input AND-OR-INVERT gate
02	Dual 2-wide, 2-input AND-OR-INVERT gate
03	Expandable 4-wide, 2-input AND-OR-INVERT gate
04	4-wide, 2-input AND-OR-INVERT gate

1.2.2 Device class. The device class shall be the product assurance level as defined in MIL-M-38510.

1.2.3 Case outline. The case outline shall be as designated in appendix C of MIL-M-38510, and as follows:

Outline letter	MIL-M-38510, appendix C, case outline
A	F-1 (14-pin, 1/4" x 1/4"), flat-package
B	F-3 (14-pin, 3/16" x 1/8"), flat-package
C	D-1 (14-pin, 1/4" x 3/4"), dual-in-line-package
D	F-2 (14-pin, 1/4" x 3/8"), flat-package

1.3 Absolute maximum ratings.

Supply voltage range	-0.5 V dc to 7.0 V dc
Input voltage range	-1.5 V dc at -12 mA to 5.5 V dc
Storage temperature range	-65°C to 150°C
Power dissipation per gate (P_D) 1/	
01, 02	51 mWdc
03, 04	68 mWdc
Lead temperature (soldering, 10 seconds)	300°C
Thermal resistance, junction-to-case (θ_{JC}):	
Cases A, B, C and D	(See MIL-M-38510 Appendix C) 2/
Junction temperature (T_J)	+175°C 3/

1/ Must withstand the added P_D due to short circuit test (e.g., I_{OS}).
2/ When a thermal resistance value is included in MIL-M-38510, Appendix C, it shall supersede the value stated herein.
3/ Maximum junction temperature shall not be exceeded except for allowable short duration burn-in screening conditions per method 5004 of MIL-STD-883.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Rome Air Development Center (RBE-2), Griffiss AFB, NY 13441, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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1.4 Recommended operating conditions.

Supply voltage(V_{CC})	4.5 V dc minimum to 5.5 V dc maximum
Minimum high-level input voltage (V_{IH})	2.0 V dc
Maximum low-level input voltage (V_{IL})	0.8 V dc
Normalized fanout (each output) 4/	10 maximum
Case operating temperature range (T_C)	-55°C to +125°C

2. APPLICABLE DOCUMENTS

2.1 Government specifications and standards. Unless otherwise specified, the following specifications and standards, of the issue listed in that issue of the Department of Defense Index of Specifications and Standards specified in the solicitation, form a part of this specification to the extent specified herein.

SPECIFICATION

MILITARY

MIL-M-38510 - Microcircuits, General Specification for.

STANDARD

MILITARY

MIL-STD-883 - Test Methods and Procedures for Microelectronics.

(Copies of specifications, standards, handbooks, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 Detail specification. The individual item requirements shall be in accordance with MIL-M-38510, and as specified herein.

3.2 Design, construction, and physical dimensions. The design, construction, and physical dimensions shall be as specified in MIL-M-38510 and herein.

3.2.1 Terminal connections and logic diagrams. The terminal connections and logic diagrams shall be as specified on figure 1.

3.2.2 Truth tables. The truth tables shall be as specified on figure 2.

3.2.3 Case outlines. Case outlines shall be in accordance with 1.2.3.

3.2.4 Schematic circuits. The schematic circuits shall be submitted to the preparing activity prior to inclusion of a manufacturer's device in this specification and shall be submitted to the qualifying activity and agent activity (DESC-ECS), as a prerequisite for qualification. All qualified manufacturers' schematics shall be maintained by the agent activity and are available upon request.

3.3 Lead material and finish. The lead material and finish shall be in accordance with MIL-M-38510. (see 6.4)

4/ The device will fanout in both high and low levels to the specified number of inputs of the same device type as that being tested.

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3.4 Electrical performance characteristics. Unless otherwise specified, the electrical performance characteristics are as specified in table I, and apply over the full recommended case operating temperature range.

TABLE I. Electrical performance characteristics.

Test	Symbol	Conditions $-55°C \leq T_C \leq +125°C$	Limits		Units
			Min	Max	
High-level output voltage	V_{OH}	V_{CC} = 4.5 V, V_{IN} = 0.8 V I_{OH} = -400 µA 1/	2.4	---	V
Low-level output voltage	V_{OL}	V_{CC} = 4.5 V, I_{OL} = 16 mA V_{IN} = 2.0 V for all inputs of gate under test 1/		0.4	V
Input clamp voltage	V_{IC}	V_{CC} = 4.5 V, I_{IN} = -12 mA T_C = 25°C		-1.5	V
High-level input current	I_{IH1}	V_{CC} = 5.5 V, V_{IN} = 2.4 V 2/		40	µA
High-level input current	I_{IH2}	V_{CC} = 5.5 V, V_{IN} = 5.5 V 2/		100	µA
Low-level input current	I_{IL}	V_{CC} = 5.5 V, V_{IN} = 0.4 V 1/	-0.7	-1.6	mA
Short-circuit output current	I_{OS}	V_{CC} = 5.5 V 2/ 3/	-20	-55	mA
High-level supply current	I_{CCH}	V_{CC} = 5.5 V 2/ V_{IN} = 0 V		7.2	mA
Low-level supply current	I_{CCL}	V_{CC} = 5.5 V 1/ 01, 02, V_{IN} = 5.5 V 03, 04		14 9.5	mA mA
Propagation delay time, high-to-low level	t_{PHL}	C_L = 50 pF, R_L = 390 Ω	3	24	ns
Propagation delay time, low-to-high level	t_{PLH}	C_L = 50 pF, R_L = 390 Ω	3	30	ns
Expander current Types 01 and 03	I_X	V_X = 0.4 V, I_{OL} = 16 mA	---	2.9	mA
Base emitter voltage	V_{BE}	I_X = 0.41 mA, I_{OL} = 16 mA	---	1.1	V

1/ All unspecified inputs at 5.5 volts.
2/ All unspecified inputs grounded.
3/ Not more than one output should be shorted at a time.

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3.5 Electrical test requirements. The electrical test requirements for each device class shall be the subgroups specified in table II. The electrical tests for each subgroup are described in table III.

TABLE II. Electrical test requirements.

MIL-STD-883 test requirements	Subgroups (see table III)	
	Class S devices	Class B devices
Interim electrical parameters (pre burn-in) (method 5004)	1	1
Final electrical test parameters (method 5004)	1*,2,3,9	1*,2,3,9
Group A test requirements (method 5005)	1,2,3,9 10,11	1,2,3,9
Group B test requirements (method 5005) subgroup 5	1,2,3,9 10,11	N/A
Groups C and D end-point electrical parameters (method 5005)	1,2,3	1,2,3
Additional electrical subgroups for group C periodic inspections	None	10,11

*PDA applies to subgroup 1 (see 4.2(d)).

3.6 Marking. Marking shall be in accordance with MIL-M-38510.

3.7 Microcircuit group assignment. The devices covered by this specification shall be in microcircuit group number 1 (see MIL-M-38510, appendix E).

4. QUALITY ASSURANCE PROVISIONS

4.1 Sampling and inspection. Sampling and inspection procedures shall be in accordance with MIL-M-38510 and method 5005 of MIL-STD-883, except as modified herein.

4.2 Screening. Screening shall be in accordance with method 5004 of MIL-STD-883, and shall be conducted on all devices prior to quality conformance inspection. The following additional criteria shall apply:

a. Burn-in test (Method 1015 of MIL-STD-883).

(1) Test condition D or E, using the circuit shown on figure 4, or equivalent.

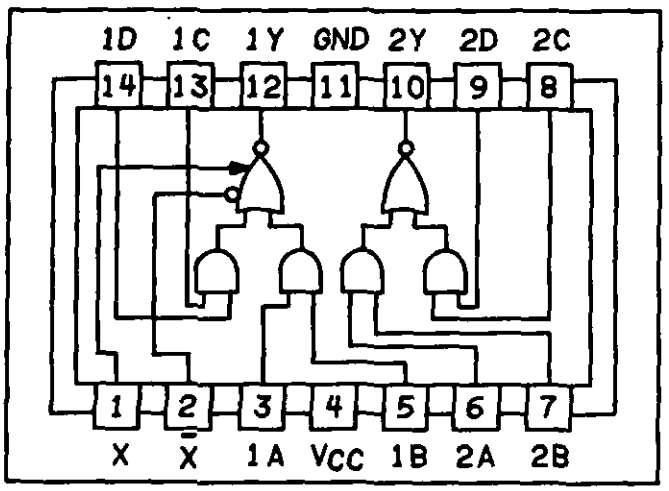
(2) T_A = 125°C minimum.

b. Reverse bias burn-in and interim electrical test in accordance with 3.1.10 of method 5004 of MIL-STD-883 shall be omitted.

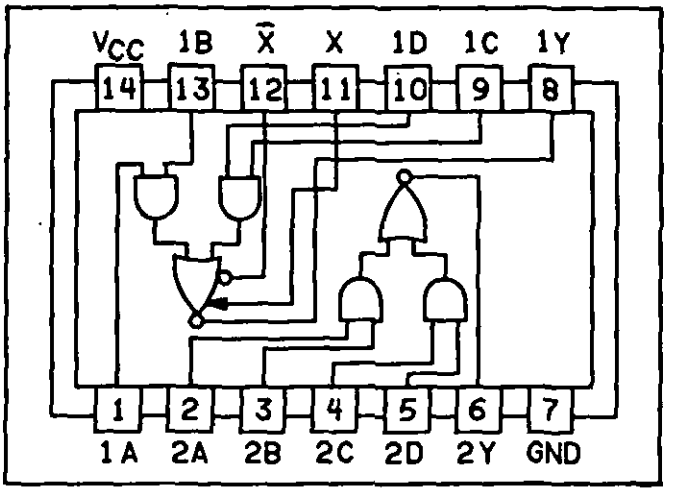
c. Interim and final electrical test parameters shall be as specified in table II herein, except interim electrical parameters test prior to burn-in is optional at the discretion of the manufacturer.

d. The percent defective allowable (PDA) shall be as specified in MIL-M-38510.

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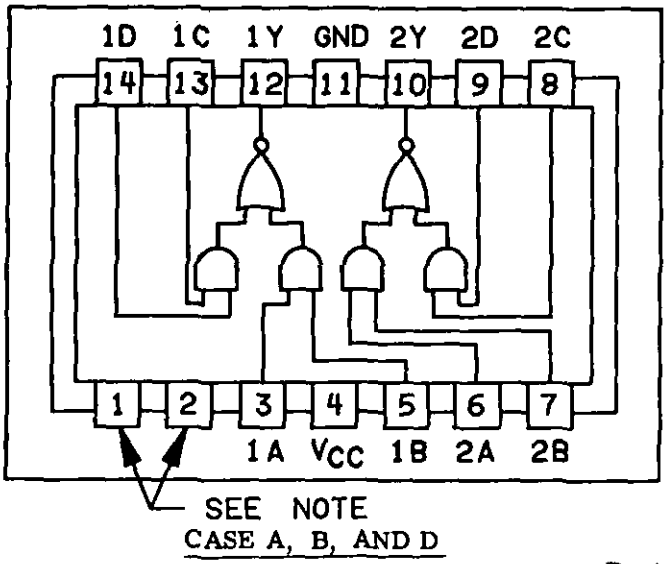


CASE A, B, AND D

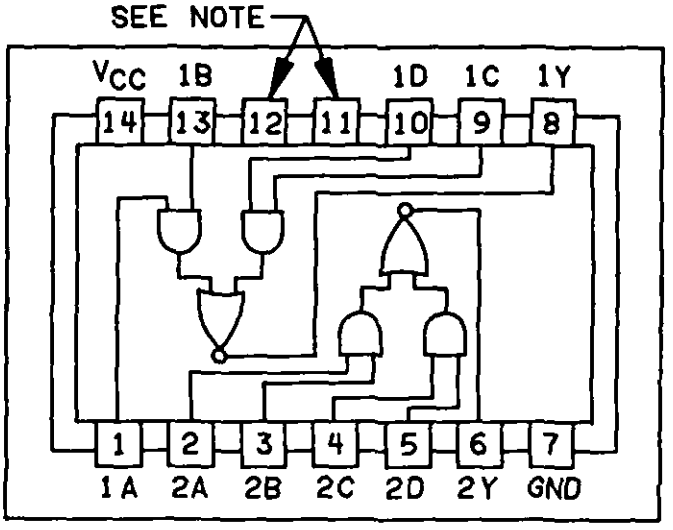


CASE C

Device Type 01



CASE A, B, AND D



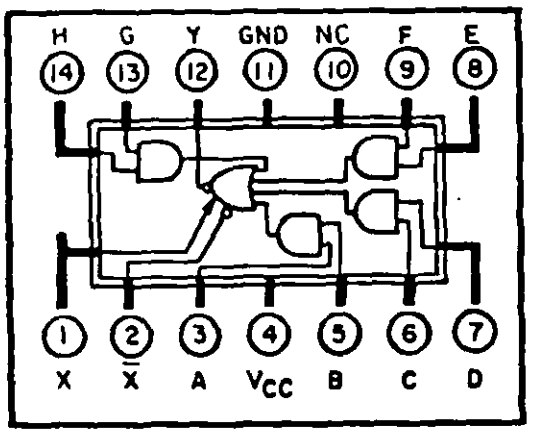
CASE C

Device Type 02

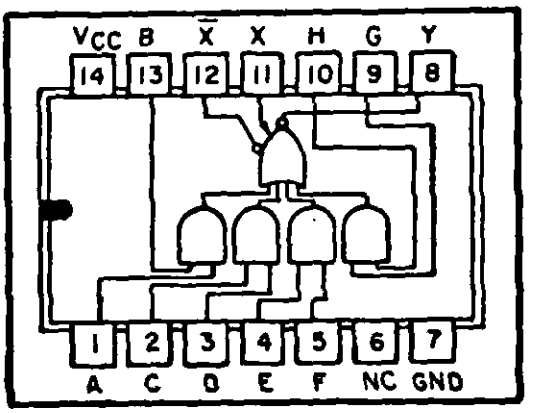
NOTE: Do not use as external tie points since they may be electrically connected internally.

FIGURE 1. Logic diagram and terminal connections (top views).

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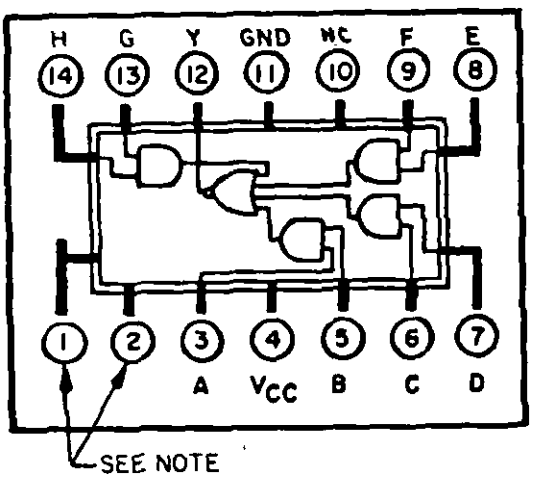


CASE A, B, AND D

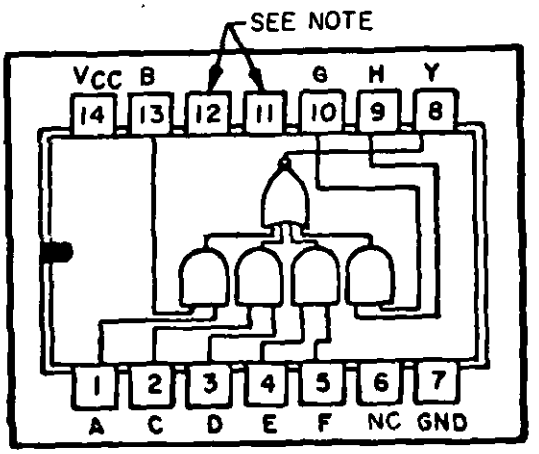


CASE C

Device Type 03



CASE A, B, AND D



CASE C

Device Type 04

NOTE: Do not use as external tie points since they may be electrically connected internally.

FIGURE 1. Logic diagram and terminal connections (top views) - Continued.

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Device Type 01

TRUTH TABLE (each gate)					
Input					Output
A	B	C	D	X	Y
H	H	*	*	*	L
*	*	H	H	*	L
*	*	*	*	H	L

* Either H or L
All other combinations of H and L at the input give H output.

Positive Logic: $Y = \overline{(AB) + (CD) + (X)}$

Device Type 02

TRUTH TABLE (each gate)				
Input				Output
A	B	C	D	Y
H	H	*	*	L
*	*	H	H	L

* Either H or L - All other combinations of H and L at the input give H output.

Positive Logic: $Y = \overline{(AB) + (CD)}$

FIGURE 2. Truth tables and logic equations.

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Device Type 03

Input									Output
A	B	C	D	E	F	G	H	X	Y
H	H	*	*	*	*	*	*	*	L
*	*	H	H	*	*	*	*	*	L
*	*	*	*	H	H	*	*	*	L
*	*	*	*	*	*	H	H	*	L
*	*	*	*	*	*	*	*	H	L

* Either H or L - All other combinations of H and L at the input give H output.

Positive logic: $Y = \overline{(AB) + (CD) + (EF) + (GH) + X}$

Device Type 04

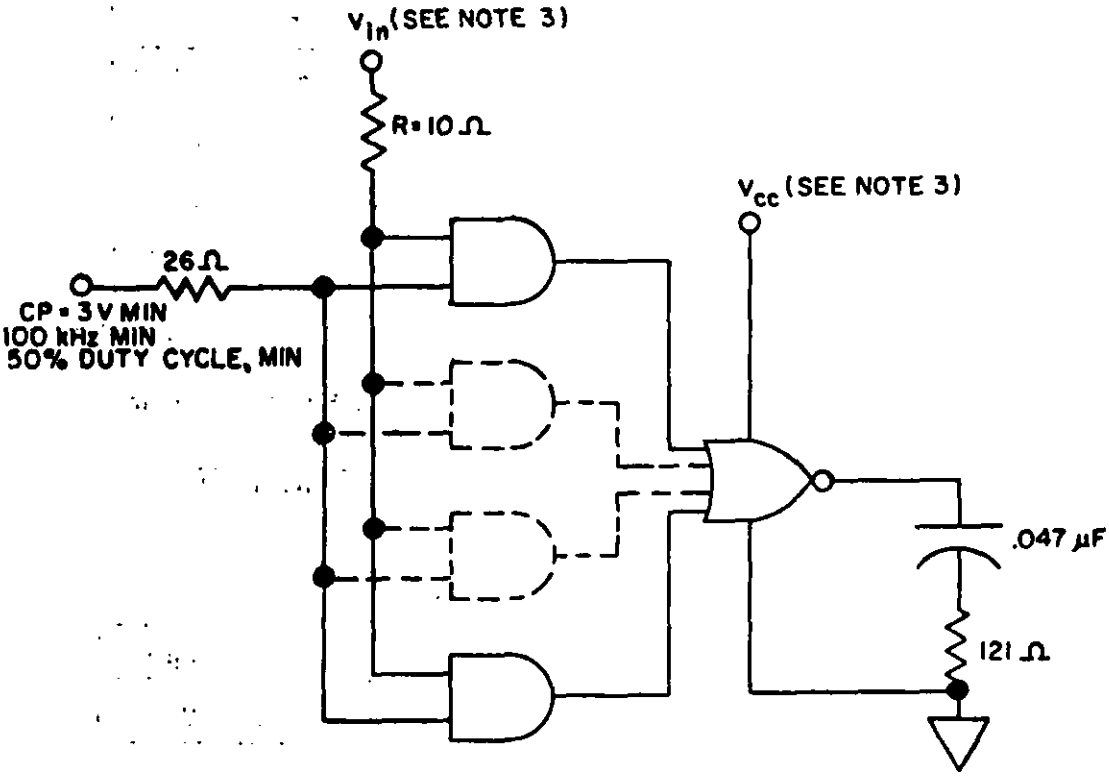
Input								Output
A	B	C	D	E	F	G	H	Y
H	H	*	*	*	*	*	*	L
*	*	H	H	*	*	*	*	L
*	*	*	*	H	H	*	*	L
*	*	*	*	*	*	H	H	L

* Either H or L - All other combinations of H and L at the input give H output.

Positive logic: $Y = \overline{(AB) + (CD) + (EF) + (GH)}$

FIGURE 2. Truth tables and logic equations - Continued.

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NOTES:
1. For -01 and -02 one gate is driven by CP and the other is driven by $\overline{CP}$.
2. Expander inputs are open.
3. V_{in} and V_{cc} = 5.0 volts, minimum, at the device terminals.

FIGURE 3. Burn-in and life test circuit.

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TABLE III. Group A inspection for device type 01.
Terminal conditions (High ≥ 2.0 V, Low ≤ 0.8 V or open)

Subgroup	Symbol	MIL-STD-883 method	Cases A,B,D / Case C / Test no.	1 / 11 / X	2 / 12 / $\overline{X}$	3 / 1 / 1A	4 / 14 / V_{CC}	5 / 13 / 1B	6 / 2 / 2A	7 / 3 / 2B	8 / 4 / 2C	9 / 5 / 2D	10 / 6 / 2Y	11 / 7 / GND	12 / 8 / 1Y	13 / 9 / 1C	14 / 10 / 1D	Measured terminal	Test limits Min	Max	Unit
1	V_{OH}	3006	1			.8 V	4.5 V	4.5 V						GND	-.4 mA	GND	GND	1Y	2.4		V
"	"	"	2			4.5 V	"	.8 V						"	"	GND	GND	1Y	"		"
"	"	"	3			GND	"	GND						"	"	.8 V	4.5 V	1Y	"		"
"	"	"	4			"	"	"						"	"	4.5 V	.8 V	1Y	"		"
"	"	"	5 1/	.15 mA	-.15 mA	"	"	"						"	"	GND	GND	1Y	"		"
"	"	"	5a 2/	0.4 V	-.15 mA	"	"	"						"	"	GND	GND	1Y	"		"
"	"	"	6				"		.8 V	4.5 V	GND	GND	-.4 mA	"				2Y	"		"
"	"	"	7				"		4.5 V	.8 V	GND	GND	"	"				2Y	"		"
"	"	"	8				"		GND	GND	.8 V	4.5 V	"	"				2Y	"		"
"	"	"	9				"		GND	GND	4.5 V	.8 V	"	"				2Y	"		"
"	V_{OL}	3007	10			2 V	"	2 V						"	16 mA	GND	GND	1Y		0.4	"
"	"	"	11			GND	"	GND						"	16 mA	2 V	2 V	1Y		"	"
"	"	"	12	.3 mA	†	GND	"	GND						"	16 mA	GND	GND	1Y		"	"
"	"	"	12a 3/	2.5 mA	1.0 V	GND	"	GND						"	16 mA	GND	GND	1Y		"	"
"	"	"	13				"		2 V	2 V	GND	GND	16 mA	"				2Y		"	"
"	"	"	14				"		GND	GND	2 V	2 V	16 mA	"				2Y		"	"
"	I_{IL}	3009	15			.4 V	5.5 V	4.5 V						"				1A	-.7	-1.6	mA
"	"	"	16			4.5 V	"	.4 V						"				1B	"	"	"
"	"	"	17				"		.4 V	4.5 V				"				2A	"	"	"
"	"	"	18				"		4.5 V	.4 V				"				2B	"	"	"
"	"	"	19				"				.4 V	4.5 V		"				2C	"	"	"
"	"	"	20				"				4.5 V	.4 V		"				2D	"	"	"
"	"	"	21				"							"		.4 V	4.5 V	1C	"	"	"
"	"	"	22				"							"		4.5 V	.4 V	1D	"	"	"
"	I_{IH1}	3010	23			2.4 V	"	GND						"				1A		40	µA
"	"	"	24			GND	"	2.4 V						"				1B		"	"
"	"	"	25				"		2.4 V	GND				"				2A		"	"
"	"	"	26				"		GND	2.4 V				"				2B		"	"
"	"	"	27				"				2.4 V	GND		"				2C		"	"
"	"	"	28				"				GND	2.4 V		"				2D		"	"
"	"	"	29				"							"		2.4 V	GND	1C		"	"
"	"	"	30				"							"		GND	2.4 V	1D		"	"

NOTES:
R_1 = 138 Ω between X and $\overline{X}$.
1/ At the manufacturer's option, the high level output voltage for the expanded inputs may be verified by use of an alternate equivalent procedure. The procedure is to omit the -400 µA current source on Y output and connect a 6K ±1% ohm resistor in parallel with V_{OH} voltmeter between the output pin and ground. The V_{OH} minimum limit is met if the resultant voltage drop is greater than 2.4 V.
2/ *The high level output voltage test for the expander input may be verified by performing either test 5 or 5a.*
3/ The low level output voltage test for the expander input may be verified by performing either test 12 or 12a.

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TABLE III. Group A inspection for device type 01. - Continued
Terminal conditions (High ≥ 2.0 V, Low ≤ 0.8 V or open)

Subgroup	Symbol	MIL-STD-883 method	Cases A,B,D Case C / Test no.	1 / 11 / X	2 / 12 / X	3 / 1 / 1A	4 / 14 / V_{CC}	5 / 13 / 1B	6 / 2 / 2A	7 / 3 / 2B	8 / 4 / 2C	9 / 5 / 2D	10 / 6 / 2Y	11 / 7 / GND	12 / 8 / 1Y	13 / 9 / 1C	14 / 10 / 1D	Measured terminal	Test limits Min	Max	Unit
1	I_{IH2}	3010	31			5.5 V	5.5 V	GND						GND				1A		100	µA
"	"	"	32			GND	"	5.5 V						"				1B		"	"
"	"	"	33				"		5.5 V	GND				"				2A		"	"
"	"	"	34				"		GND	5.5 V				"				2B		"	"
"	"	"	35				"							"				2C		"	"
"	"	"	36				"				5.5 V	GND		"				2D		"	"
"	"	"	37				"				GND	5.5 V		"		5.5 V	GND	1C		"	"
"	"	"	38				"							"		GND	5.5 V	1D		"	"
	I_{OS}	3011	39			GND	"	GND						"	GND	GND	GND	1Y	-20	-55	mA
	I_{OS}	3011	40				"		GND	GND	GND	GND	GND	"				2Y	-20	-55	"
	I_{CCL}	3005	41			5.5 V	"	5.5 V	5.5 V	5.5 V	5.5 V	5.5 V		"		5.5 V	5.5 V	V_{CC}		14	"
	I_{CCH}	3005	42			GND	"	GND	GND	GND	GND	GND		"		GND	GND	V_{CC}		7.2	"
	V_{IC}		43			-12 mA	4.5 V							"				1A		-1.5	V
"	"	"	44				"	-12 mA						"				1B		"	"
"	"	"	45				"							"		-12 mA		1C		"	"
"	"	"	46				"							"			12 mA	1D		"	"
"	"	"	47				"		-12 mA					"				2A		"	"
"	"	"	48				"			-12 mA				"				2B		"	"
"	"	"	49				"				-12 mA			"				2C		"	"
"	"	"	50				"					-12 mA		"				2D		"	"
	I_X	3/	51	-.4 V	1/	GND	"	GND	GND	GND	GND			"	16 mA	GND	GND	X		2.9	mA
	I_X	3/	51a		1.4 V	"	"	"	"	"	"			"	16 mA	"	"	X		2.58	mA
	V_X	3/	51b	1.7 mA	0.8 V	"	"	"	"	"	"			"	16 mA	"	"	X	0.62		V
	V_{BE}	2/	52	.41 mA		"	"	"	"	"	"			"	16 mA	"	"	X		1.1	V
2	Same tests, terminal conditions and limits as for subgroup 1, except T_C = +125°C and V_{IC} tests are omitted.																				
3	Same tests, terminal conditions and limits as for subgroup 1, except T_C = -55°C and V_{IC} tests are omitted.																				
9	t_{PHL}	3003 (Fig 4)	53			IN	5.0 V	2.4 V	GND	GND	GND	GND		GND	OUT	GND	GND	1A to 1Y	3	20	ns
"	"	"	54			GND	"	GND	IN	2.4 V	"	"		"		"	"	2A to 2Y	3	20	"
	t_{PLH}	"	55			IN	"	2.4 V	GND	GND	"	"		"	OUT	"	"	1A to 1Y	3	27	"
"	"	"	56			GND	"	GND	IN	2.4 V	"	"		"		"	"	2A to 2Y	3	27	"
10 T_C = +125°C	t_{PHL}	"	55			IN	"	2.4 V	GND	GND	"	"		"	OUT	"	"	1A to 1Y	3	24	"
"	"	"	56			GND	"	GND	IN	2.4 V	"	"		"		"	"	2A to 2Y	3	24	"
	t_{PLH}	"	55			IN	"	2.4 V	GND	GND	"	"		"	OUT	"	"	1A to 1Y	3	30	"
"	"	"	56			GND	"	GND	IN	2.4 V	"	"		"		"	"	2A to 2Y	3	30	"
11	Same tests, terminal conditions and limits as for subgroup 10, except T_C = -55°C.																				

NOTES: 1/ -.4 V at X referenced to $\overline{X}$ (see figure 5).
2/ X and $\overline{X}$ shorted (see figure 6).
3/ At the manufacturer's option, the expander current test may be verified by performing either 51 or 51a and 51b.

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TABLE III. Group A inspection for device type 02.
Terminal conditions (High ≥ 2.0 V, Low ≤ 0.8 V or open)

Subgroup	Symbol	MIL-STD-883 method	Cases A,B,D Case C Test no.	1 11 NC	2 12 NC	3 1 1A	4 14 V_{CC}	5 13 1B	6 2 2A	7 3 2B	8 4 2C	9 5 2D	10 6 2Y	11 7 GND	12 8 1Y	13 9 1C	14 10 1D	Measured terminal	Test limits Min	Max	Unit
1	V_{OH}	3006	1			.8 V	4.5 V	4.5 V						GND	-.4 mA	GND	GND	1Y	2.4		V
	"	"	2			4.5 V	"	.8 V						"	"	GND	GND	1Y	"		"
	"	"	3			GND	"	GND						"	"	.8 V	4.5 V	1Y	"		"
	"	"	4			GND	"	GND						"	"	4.5 V	.8 V	1Y	"		"
	"	"	5				"		.8 V	4.5 V	GND	GND	-.4 mA	"				2Y	"		"
	"	"	6				"		4.5 V	.8 V	GND	GND	"	"				2Y	"		"
	"	"	7				"		GND	GND	.8 V	4.5 V	"	"				2Y	"		"
	"	"	8				"		GND	GND	4.5 V	.8 V	"	"				2Y	"		"
	V_{OL}	3007	9			2 V	"	2 V						"	16 mA	GND	GND	1Y		.4	V
	"	"	10			GND	"	GND						"	16 mA	2 V	2 V	1Y		"	"
	"	"	11				"		2 V	2 V	GND	GND	16 mA	"				2Y		"	"
	"	"	12				"		GND	GND	2 V	2 V	16 mA	"				2Y		"	"
	I_{IL}	3009	13			.4 V	5.5 V	4.5 V						"				1A	-.7	-1.6	mA
	"	"	14			4.5 V	"	.4 V						"				1B	"	"	"
	"	"	15				"		.4 V	4.5 V				"				2A	"	"	"
	"	"	16				"		4.5 V	.4 V				"				2B	"	"	"
	"	"	17				"				.4 V	4.5 V		"				2C	"	"	"
	"	"	18				"				4.5 V	.4 V		"				2D	"	"	"
	"	"	19				"							"		.4 V	4.5 V	1C	"	"	"
	"	"	20				"							"		4.5 V	.4 V	1D	"	"	"
	I_{IH1}	3010	21			2.4 V	"	GND						"				1A		40	uA
	"	"	22			GND	"	2.4 V						"				1B		"	"
	"	"	23				"		2.4 V	GND				"				2A		"	"
	"	"	24				"		GND	2.4 V				"				2B		"	"
	"	"	25				"				2.4 V	GND		"				2C		"	"
	"	"	26				"				GND	2.4 V		"				2D		"	"
	"	"	27				"							"		2.4 V	GND	1C		40	"
	"	"	28				"							"		GND	2.4 V	1D		40	"
	I_{IH2}	"	29			5.5 V	"	GND						"				1A		100	"
	"	"	30			GND	"	5.5 V						"				1B		"	"
	"	"	31				"		5.5 V	GND				"				2A		"	"
	"	"	32				"		GND	5.5 V				"				2B		"	"
	"	"	33				"				5.5 V	GND		"				2C		"	"
	"	"	34				"				GND	5.5 V		"				2D		"	"
	"	"	35				"							"		5.5 V	GND	1C		"	"
	"	"	36				"							"		GND	5.5 V	1D		"	"

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TABLE III. Group A inspection for device type 02. - Continued
Terminal conditions (High ≥ 2.0 V, Low ≤ 0.8 V or open)

Subgroup	Symbol	MIL-STD-883 method	Cases A,B,D	1	2	3	4	5	6	7	8	9	10	11	12	13	14	Measured terminal	Test limits		
			Case C	11	12	1	14	13	2	3	4	5	6	7	8	9	10		Min	Max	Unit
			Test no.	NC	NC	1A	V_{CC}	1B	2A	2B	2C	2D	2Y	GND	1Y	1C	1D				
1	I_{OS}	3011	37			GND	5.5 V	GND						GND	GND	GND	GND	1Y	-20	-55	mA
"	I_{OS}	3011	38				"		GND	GND	GND	GND	GND	"				2Y	-20	-55	"
"	I_{CCL}	3005	39			5.5 V	"	5.5 V	5.5 V	5.5 V	5.5 V	5.5 V		"		5.5 V	5.5 V	V_{CC}		14	"
"	I_{CCH}	3005	40			GND	"	GND	GND	GND	GND	GND		"		GND	GND	V_{CC}		7.2	"
"	V_{IC}		41			-12 mA	4.5 V							"				1A		-1.5	V
"	"		42				"	-12 mA						"				1B		"	"
"	"		43				"							"		-12 mA		1C		"	"
"	"		44				"							"			12 mA	1D		"	"
"	"		45				"		-12 mA					"				2A		"	"
"	"		46				"			-12 mA				"				2B		"	"
"	"		47				"				-12 mA			"				2C		"	"
"	"		48				"					-12 mA		"				2D		"	"
2	Same tests, terminal conditions and limits as for subgroup 1, except T_C = +125°C and V_{IC} tests are omitted.																				
3	Same tests, terminal conditions and limits as for subgroup 1, except T_C = -55°C and V_{IC} tests are omitted.																				
9	t_{PHL}	3003 (Fig 4)	49			IN	5.0 V	2.4 V	GND	GND	GND	GND		GND	OUT	GND	GND	1A to 1Y	3	20	ns
"		"	50			GND	"	GND	IN	2.4 V	"	"	OUT			"	"	2A to 2Y	"	20	"
"	t_{PLH}	"	51			IN	"	2.4 V	GND	GND	"	"		"	OUT	"	"	1A to 1Y	"	27	"
"		"	52			GND	"	GND	IN	2.4 V	"	"	OUT	"		"	"	2A to 2Y	"	27	"
10 T_C = +125°C	t_{PHL}	"	53			IN	"	2.4 V	GND	GND	"	"		"	OUT	"	"	1A to 1Y	"	24	"
"	"	"	54			GND	"	GND	IN	2.4 V	"	"	OUT	"		"	"	2A to 2Y	"	24	"
"	t_{PLH}	"	55			IN	"	2.4 V	GND	GND	"	"		"	OUT	"	"	1A to 1Y	"	30	"
"	"	"	56			GND	"	GND	IN	2.4 V	"	"	OUT	"		"	"	2A to 2Y	"	30	"
11	Same tests, terminal conditions and limits as for subgroup 10, except T_C = -55°C.																				

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TABLE III. Group A inspection for device type 03.
Terminal conditions (High ≥ 2.0 V, Low ≤ 0.8 V or open)

Subgroup	Symbol	MIL-STD-883 method	Cases A,8,D / Case C / Test no.	1 / 11 / X	2 / 12 / $\overline{X}$	3 / 1 / A	4 / 14 / V_{CC}	5 / 13 / B	6 / 2 / C	7 / 3 / D	8 / 4 / E	9 / 5 / F	10 / 6 / NC	11 / 7 / GND	12 / 8 / Y	13 / 9 / G	14 / 10 / H	Measured terminal	Test limits Min	Max	Unit
1 T_C = +25°C "	V_{OH}	3006	1			0.8 V	4.5 V	4.5 V	0.8 V	4.5 V	0.8 V	4.5 V		GND	-.4 mA	0.8 V	4.5 V	Y	2.4		V
"	"	"	2			4.5 V	"	0.8 V	4.5 V	0.8 V	4.5 V	0.8 V		"	"	4.5 V	0.8 V	"	"		"
"	"	"	3 1/	.15 mA	-.15 mA	GND	"	GND	GND	GND	GND	GND		"	"	GND	GND	"	"		"
			3a 2/	0.4 V	-.15 mA	GND	"	GND	GND	GND	GND	GND		"	"	GND	GND	"	"		"
"	V_{OL}	3007	4			2 V	"	2 V	GND	GND	GND	GND		"	16 mA	GND	GND	"		0.4	"
"	"	"	5			GND	"	GND	2 V	2 V	GND	GND		"	"	GND	GND	"		"	"
"	"	"	6			"	"	"	GND	GND	2 V	2 V		"	"	GND	GND	"		"	"
"	"	"	7			"	"	"	"	"	GND	GND		"	"	2.7 V	2.7 V	"		"	"
"	"	"	8	.3 mA	3/	"	"	"	"	"	GND	GND		"	"	GND	GND	"		"	"
"	"	"	8a 4/	2.5 mA	1.0 V	"	"	"	"	"	GND	GND		"	"	GND	GND	"		"	"
"	I_{IL}	3009	9			0.4 V	5.5 V	4.5 V						"				A	-.7	-1.6	mA
"	"	"	10			4.5 V	"	0.4 V						"				B	"	"	"
"	"	"	11				"		0.4 V	4.5 V				"				C	"	"	"
"	"	"	12				"		4.5 V	0.4 V				"				D	"	"	"
"	"	"	13				"				0.4 V	4.5 V		"				E	"	"	"
"	"	"	14				"				4.5 V	0.4 V		"				F	"	"	"
"	"	"	15				"							"		0.4 V	4.5 V	G	"	"	"
"	"	"	16				"							"		4.5 V	0.4 V	H	"	"	"
"	I_{IH1}	3010	17			2.4 V	"	GND						"				A		40	uA
"	"	"	18			GND	"	2.4 V						"				B		"	"
"	"	"	19				"		2.4 V	GND				"				C		"	"
"	"	"	20				"		GND	2.4 V				"				D		"	"
"	"	"	21				"				2.4 V	GND		"				E		"	"
"	"	"	22				"				GND	2.4 V		"				F		"	"
"	"	"	23				"							"		2.4 V	GND	G		"	"
"	"	"	24				"							"		GND	2.4 V	H		"	"
"	I_{IH2}	3010	25			5.5 V	"	GND						"				A		100	uA
"	"	"	26			GND	"	5.5 V						"				B		"	"
"	"	"	27				"		5.5 V	GND				"				C		"	"
"	"	"	28				"		GND	5.5 V				"				D		"	"
"	"	"	29				"				5.5 V	GND		"				E		"	"
"	"	"	30				"				GND	5.5 V		"				F		"	"
"	"	"	31				"							"		5.5 V	GND	G		"	"
"	"	"	32				"							"		GND	5.5 V	H		"	"
1	I_{OS}	3011	33			GND	"	GND	GND	GND	GND	GND		"	GND	GND	GND	Y	-20	-55	mA
"	I_{CCL}	3005	34			5.5 V	"	5.5 V	5.5 V	5.5 V	5.5 V	5.5 V		"		5.5 V	5.5 V	V_{CC}		9.5	mA
"	I_{CCH}	3005	35			GND	"	GND	GND	GND	GND	GND		"		GND	GND	V_{CC}		7.2	mA

1/ At the manufacturer's option, the high level output voltage for the expanded inputs may be verified by use of an alternate equivalent procedure. The procedure is to omit the -400 uA current source on Y output and connect a 6K ±1% ohm resistor in parallel with V_{OH} voltmeter between the output pin and ground. The V_{OH} minimum limit is met if the resultant voltage drop is greater than 2.4 V.
2/ The high level output voltage test for the expander input may be verified by performing either test 3 or 3a.
3/ R_1 = 138 Ω between X and $\overline{X}$.
4/ The low level output voltage test for the expander input may be verified by performing either test 8 or 8a.

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TABLE III. Group A inspection for device type 03. - Continued
Terminal conditions (High ≥ 2.0 V, Low ≤ 0.8 V or open)

Subgroup	Symbol	MIL-STD-883 method	Cases A,B,D	1	2	3	4	5	6	7	8	9	10	11	12	13	14	Measured terminal	Test limits		
			Case C	11	12	1	14	13	2	3	4	5	6	7	8	9	10		Min	Max	Unit
			Test no.	X	$\overline{X}$	A	V_{CC}	B	C	D	E	F	NC	GND	Y	G	H				
T_C = +25°C	V_{IC}	3005	36			-12 mA	4.5 V							GND				A		-1.5	V
"	"		37			"	"	-12 mA						"				B		"	"
"	"		38				"		-12 mA					"				C		"	"
"	"		39				"			-12 mA				"				D		"	"
"	"		40				"				-12 mA			"				E		"	"
"	"		41				"					-12 mA		"				F		"	"
"	"		42				"							"		-12 mA		G		"	"
"	"		43				"							"			-12 mA	H		"	"
"	I_X	3/	44	-.4 V	1/	GND	"	GND	GND	GND	GND	GND		"	16 mA	GND	GND	$\overline{X}$		2.9	mA
"	I_X	3/	44a		1.4 V	"	"	"	"	"	"	"		"	"	"	"	$\overline{X}$		2.58	mA
"	V_X	3/	44b	1.7 mA	0.8 V	"	"	"	"	"	"	"		"	"	"	"	X	0.62		V
"	V_{BE}		45	.41 mA	2/	"	"	"	"	"	"	"		"	"	"	"	X		1.1	V
2	Same tests, terminal conditions and limits as for subgroup 1, except T_C = +125°C and V_{IC} tests are omitted.																				
3	Same tests, terminal conditions and limits as for subgroup 1, except T_C = -55°C and V_{IC} tests are omitted.																				
9	t_{PHL}	3003 (Fig 4)	46			IN	5.0 V	2.4 V	GND	GND	GND	GND		GND	OUT	GND	GND	A to Y	3	20	ns
	t_{PLH}	"	47			"	"	"	"	"	"	"		"	"	"	"	A to Y	"	27	"
10 T_C = +125°C	t_{PHL}	"	48			"	"	"	"	"	"	"		"	"	"	"	A to Y	"	24	"
	t_{PLH}	"	49			"	"	"	"	"	"	"		"	"	"	"	A to Y	"	30	"
11	Same tests, terminal conditions and limits as for subgroup 10, except T_C = -55°C.																				

1/ -.4 V at X referenced to $\overline{X}$ (see figure 5).
2/ X and $\overline{X}$ shorted (see figure 6).
3/ At the manufacturer's option, the expander current test may be verified by performing either 44 or 44a and 44b.

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TABLE III. Group A inspection for device type 04.
Terminal conditions (High ≥ 2.0 V, Low ≤ 0.8 V or open)

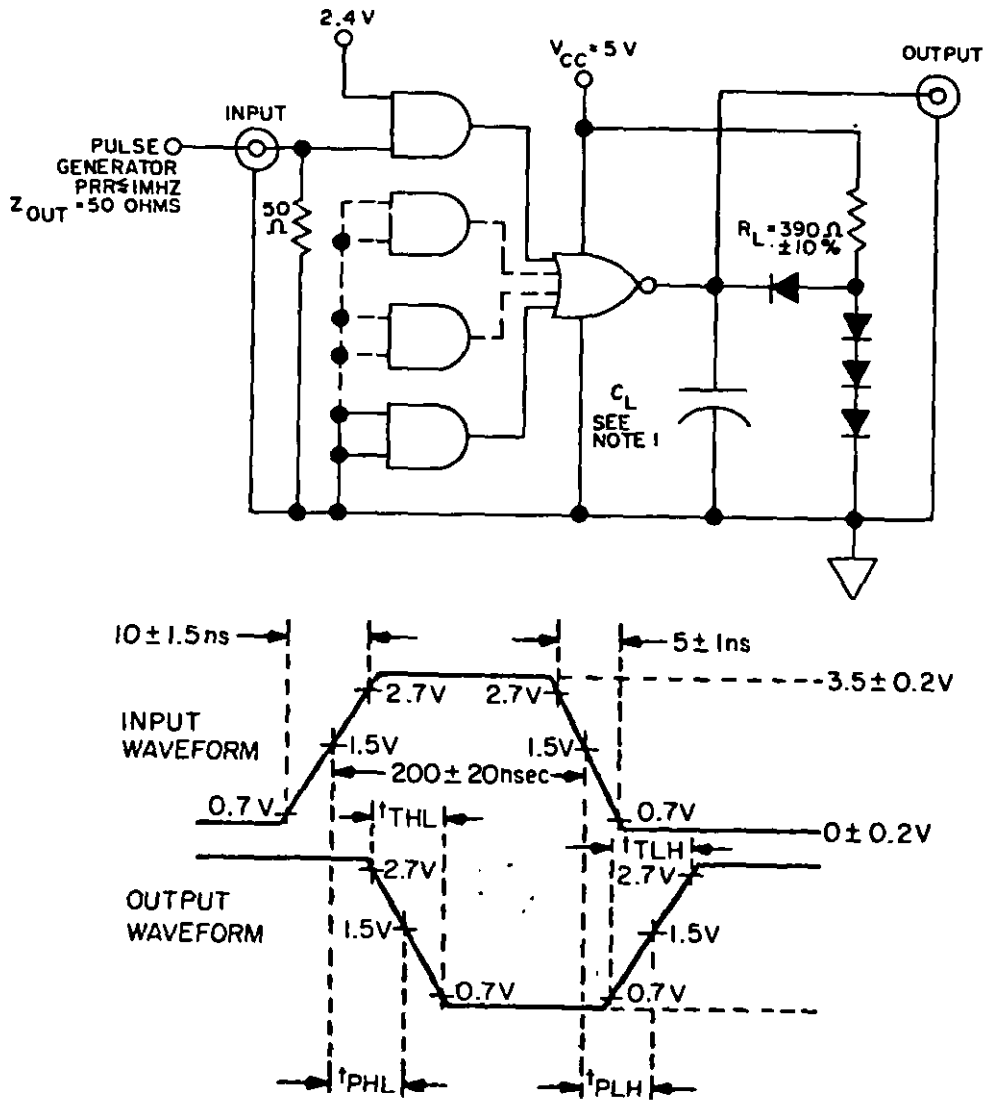
Subgroup	Symbol	MIL-STD-883 method	Cases A,B,D Case C Test no.	1 11 NC	2 12 NC	3 1 A	4 14 V_{CC}	5 13 B	6 2 C	7 3 D	8 4 E	9 5 F	10 6 NC	11 7 GND	12 8 Y	13 9 G	14 10 H	Measured terminal	Test limits Min	Max	Unit
1	V_{OH}	3006	1			.8 V	4.5 V	4.5 V	.8 V	4.5 V	.8 V	4.5 V		GND	-.4 mA	.8 V	4.5 V	Y	2.4		V
"	V_{OH}	3006	2			4.5 V	"	.8 V	4.5 V	.8 V	4.5 V	.8 V		"	-.4 mA	4.5 V	.8 V	Y	2.4		"
"	V_{OL}	3007	3			2 V	"	2 V	GND	GND	GND	GND		"	16 mA	GND	GND	Y		.4	"
"	"	"	4			GND	"	GND	2 V	2 V	GND	GND		"	"	GND	GND	Y		"	"
"	"	"	5			"	"	"	GND	GND	2 V	2 V		"	"	GND	GND	Y		"	"
"	"	"	6			"	"	"	GND	GND	GND	GND		"	"	2 V	2 V	Y		"	"
"	I_{IL}	3009	7			.4 V	5.5 V	4.5 V						"				A	-.7	-1.6	mA
"	"	"	8			4.5 V	"	.4 V						"				B	"	"	"
"	"	"	9				"		.4 V	4.5 V				"				C	"	"	"
"	"	"	10				"		4.5 V	.4 V				"				D	"	"	"
"	"	"	11				"				.4 V	4.5 V		"				E	"	"	"
"	"	"	12				"				4.5 V	.4 V		"				F	"	"	"
"	"	"	13				"							"		.4 V	4.5 V	G	"	"	"
"	"	"	14				"							"		4.5 V	.4 V	H	"	"	"
"	I_{IH1}	3010	15			2.4 V	"	GND						"				A		40	uA
"	"	"	16			GND	"	2.4 V						"				B		"	"
"	"	"	17				"		2.4 V	GND				"				C		"	"
"	"	"	18				"		GND	2.4 V				"				D		"	"
"	"	"	19				"				2.4 V	GND		"				E		"	"
"	"	"	20				"				GND	2.4 V		"				F		"	"
"	"	"	21				"							"		2.4 V	GND	G		"	"
"	"	"	22				"							"		GND	2.4 V	H		"	"
"	I_{IH2}	3010	23			5.5 V	"	GND						"				A		100	"
"	"	"	24			GND	"	5.5 V						"				B		"	"
"	"	"	25				"		5.5 V	GND				"				C		"	"
"	"	"	26				"		GND	5.5 V				"				D		"	"
"	"	"	27				"				5.5 V	GND		"				E		"	"
"	"	"	28				"				GND	5.5 V		"				F		"	"
"	"	"	29				"							"		5.5 V	GND	G		"	"
"	"	"	30				"							"		GND	5.5 V	H		"	"
"	I_{OS}	3011	31			GND	"	GND	GND	GND	GND	GND		"	GND	GND	GND	Y	-20	-55	mA
"	I_{CCL}	3005	32			5.5 V	"	5.5 V	5.5 V	5.5 V	5.5 V	5.5 V		"		5.5 V	5.5 V	V_{CC}		9.5	mA
"	I_{CCH}	3005	33			GND	"	GND	GND	GND	GND	GND		"		GND	GND	V_{CC}		7.2	mA

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TABLE III. Group A inspection for device type 04. - Continued
Terminal conditions (High ≥ 2.0 V, Low ≤ 0.8 V or open)

Subgroup	Symbol	MIL-STD-883 method	Cases A,B,D / Case C / Test no.	1 / 11 / NC	2 / 12 / NC	3 / 1 / A	4 / 14 / V_{CC}	5 / 13 / B	6 / 2 / C	7 / 3 / D	8 / 4 / E	9 / 5 / F	10 / 6 / NC	11 / 7 / GND	12 / 8 / Y	13 / 9 / G	14 / 10 / H	Measured terminal	Test limits Min	Max	Unit
1	V_{IC}		34			-12 mA	4.5 V							"				A		-1.5	V
"	"		35			"	"	-12 mA						"				B		"	"
"	"		36			"	"	"	-12 mA					"				C		"	"
"	"		37			"	"	"	"	-12 mA				"				D		"	"
"	"		38			"	"	"	"	"	-12 mA			"				E		"	"
"	"		39			"	"	"	"	"	"	-12 mA		"				F		"	"
"	"		40			"	"	"	"	"	"	"		"		-12 mA		G		"	"
"	"		41			"	"	"	"	"	"	"		"		"	-12 mA	H		"	"
2	Same tests, terminal conditions and limits as for subgroup 1, except T_C = +125°C and V_{IC} tests are omitted.																				
3	Same tests, terminal conditions and limits as for subgroup 1, except T_C = -55°C and V_{IC} tests are omitted.																				
9	t_{PHL}	3003	42			IN	5.0 V	2.4 V	GND	GND	GND	GND		GND	OUT	GND	GND	A to Y	3	20	ns
	t_{PLH}	(Fig 4)	43			"	"	"	"	"	"	"		"	"	"	"	A to Y	"	27	"
10 T_C = +125°C	t_{PHL}	"	44			"	"	"	"	"	"	"		"	"	"	"	A to Y	"	24	"
	t_{PLH}	"	45			"	"	"	"	"	"	"		"	"	"	"	A to Y	"	30	"
11	Same tests, terminal conditions and limits as for subgroup 10, except T_C = -55°C.																				

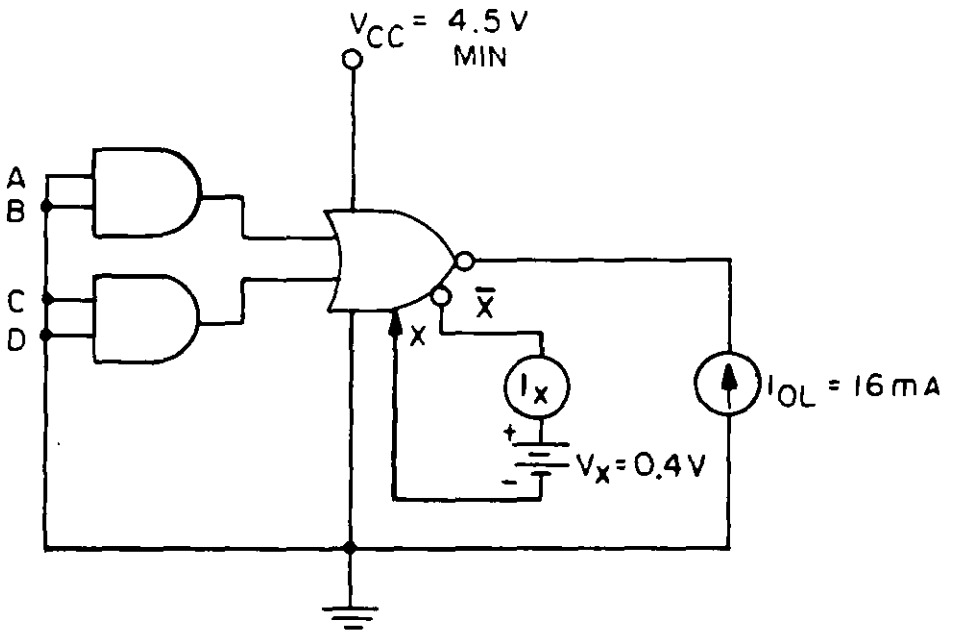
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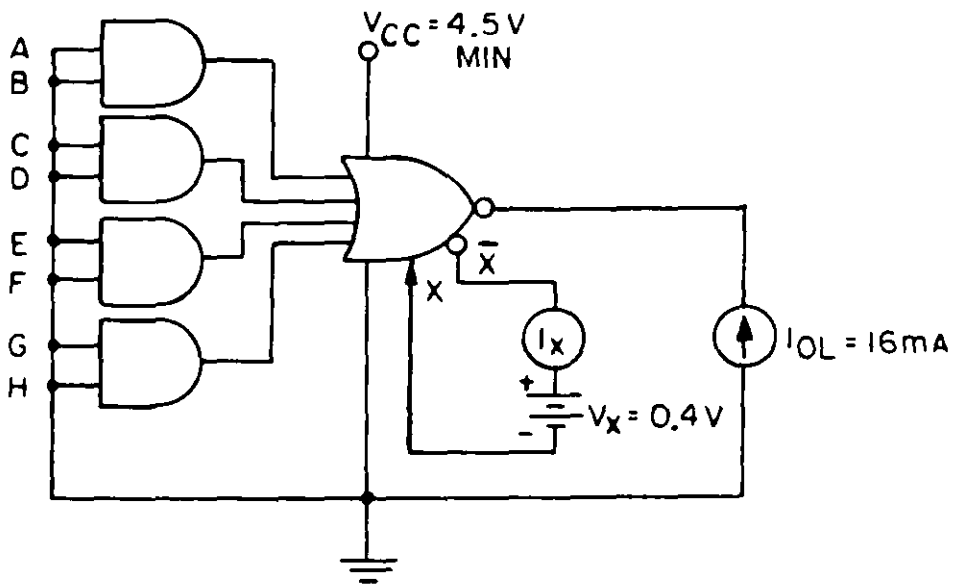
NOTES:
1. C_L = 50pf including scope probe, wiring, and stray capacitance, without package in test fixture.
2. Voltage measurements are to be made with respect to network ground terminal.
3. All diodes are 1N3064 or equivalent.

FIGURE 4. Switching time test circuit.

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Device type 01

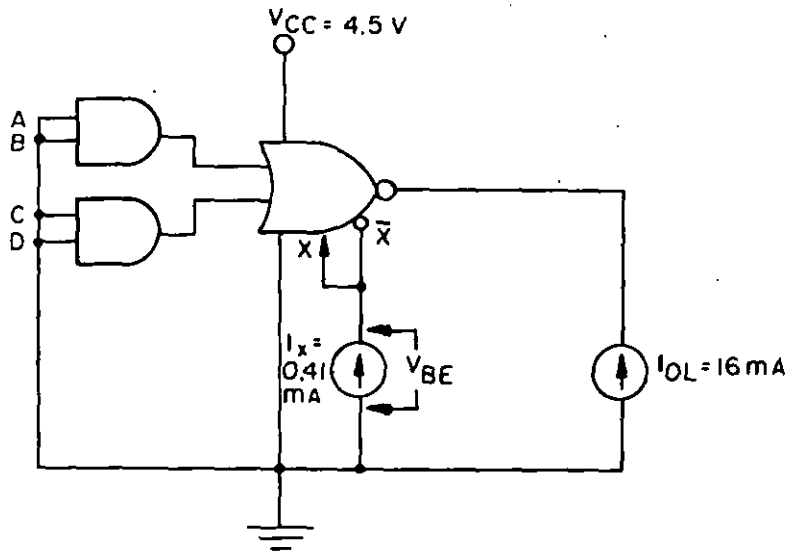


Device type 03

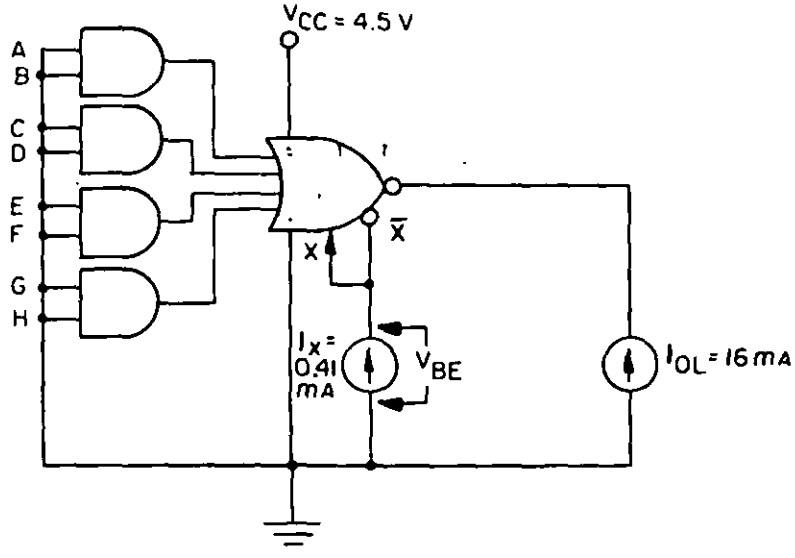
NOTE:
At the manufacturer's option, the expander test current limit may be verified by the use of an alternate equivalent procedure. The V_X voltage source and the I_X meter may be replaced with a 138Ω resistor in parallel with a voltmeter between X and $\overline{X}$. When the applicable conditions are applied, the resultant voltage drop across the resistor is measured. The I_X limit of 2.9mA is met if the resultant voltage does not exceed 0.4V.

FIGURE 5. Expander current test circuit for device types 01 and 03.

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Device type 01



Device type 03

FIGURE 6. Base emitter voltage test circuit for device types 01 and 03.

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4.3 Qualification inspection. Qualification inspection shall be in accordance with MIL-M-38510. Inspections to be performed shall be those specified in method 5005 of MIL-STD-883 and herein for groups A, B, C, and D inspections (see 4.4.1 , 4.4.2, and 4.4.3).

4.4 Quality conformance inspection. Quality conformance inspection shall be in accordance with MIL-M-38510 and as specified herein. Inspections to be performed shall be those specified in method 5005 of MIL-STD-883 and herein for groups A, B, C, and D inspections (see 4.4.1 through 4.4.4).

4.4.1 Group A inspection. Group A inspection shall be in accordance with table I of method 5005 of MIL-STD-883 and as follows:

a. Test shall be as specified in table II.

b. Subgroups 4, 5, 6, 7, and 8 shall be omitted.

4.4.2 Group B inspection. Group B inspection shall be in accordance with table II of method 5005 of MIL-STD-883. Electrical parameters shall be as specified in table II herein.

4.4.3 Group C and D inspection. Groups C and D inspection shall be in accordance with table III and IV of method 5005 of MIL-STD-883 and as follows:

a. End-point electrical parameters shall be as specified in table II.

b. Subgroups 3 and 4 shall be added to group C inspection requirements for class B devices and shall consist of the tests, conditions, and limits specified for subgroups 10 and 11 of group A.

c. Operating life test (Method 1005 of MIL-STD-883) conditions:

(1) Test condition D or E, using the circuit shown on figure 3, or equivalent.

(2) T_A = 125°C minimum.

(3) Test duration - 1,000 hours, except as permitted by method 1005 of MIL-STD-883.

4.5 Methods of inspection. Methods of inspection shall be specified as follows:

4.5.1 Voltage and current. All voltages given are referenced to the microcircuit ground terminal. Currents given are conventional and positive when flowing into the referenced terminal.

5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with MIL-M-38510.

6. NOTES

6.1 Intended use. Microcircuits conforming to this specification are intended for use for Government microcircuit applications (original equipment) and logisitic purposes.

6.2 Ordering data. The acquisition document should specify the following:

a. Complete part number (see 1.2).

b. Requirements for delivery of one copy of the quality conformance inspection data pertinent to the device inspection lot to be supplied with each shipment by the device manufacturer, if applicable.

c. Requirements for certificate of compliance, if applicable.

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d. Requirements for notification of change of product or process to contracting activity in addition to notification to the qualifying activity, if applicable.

e. Requirements for failure analysis (including required test condition of method 5003 of MIL-STD-883), corrective action and reporting of results, if applicable.

f. Requirements for product assurance options.

g. Requirements for special carriers, lead lengths, or lead forming, if applicable. These requirements shall not affect the part number. Unless otherwise specified, these requirements shall not apply to direct purchase by or direct shipment to the Government.

h. Requirements for "JAN" marking.

6.3 Abbreviations, symbols, and definitions. The abbreviations, symbols, and definitions used herein are defined in MIL-M-38510, MIL-STD-1331, and as follows:

GND	Ground zero voltage potential.
V_{IN}	Voltage level at an input terminal.
V_{IC}	Input clamp voltage.

6.4 Logistic support. Lead materials and finishes (see 3.3) are interchangeable. Unless otherwise specified, microcircuits acquired for Government logistic support will be acquired to device class B (see 1.2.2), lead material and finish C (see 3.3), and case outline A (see 1.2.3). There is no substitute for case outline C. Longer length leads and lead forming shall not affect the part number.

6.5 Substitutability. The cross-reference information below is presented for the convenience of users. Microcircuits covered by this specification will functionally replace the listed generic-industry type. Generic-industry microcircuit types may not have equivalent operational performance characteristics across military temperature ranges or reliability factors equivalent to MIL-M-38510 device types and may have slight physical variations in relation to case size. The presence of this information shall not be deemed as permitting substitution of generic-industry types for MIL-M-38510 types or as a waiver of any of the provisions of MIL-M-38510.

Device type	Commercial type
01	5450 (Circuit A)
01	S5450 (Circuit B)
02	5451 (Circuit A)
02	S5451 (Circuit B)
03	5453 (Circuit A)
03	S5453 (Circuit B)
04	5454 (Circuit A)
04	S5454 (Circuit B)

6.6 Manufacturers' designators. Manufacturers' circuits included in this specification are designated with an "X" as shown in table IV herein.

TABLE IV. Manufacturers' designations.

Device type	Circuit		
	A	B	C
	National Semiconductor Corp.	Texas Instruments	Motorola Inc.
01	X	X	X
02	X	X	X
03	X	X	X
04	X	X	X

6.7 In lieu of documents. MIL-M-0038510/5B(19) dated 1 December 1975 was issued as an "In lieu of" document for MIL-M-38510/5A dated 6 December 1971. This revision, MIL-M-38510/5C dated 20 MAY 1985 supersedes MIL-M-38510/5B dated 7 January 1985, MIL-M-0038510/5B(19) dated 1 December 1975 and MIL-M-38510/5A dated 6 December 1971.

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6.8 Changes from previous issue: Asterisks are not used in this revision to indicate changes from the previous issue due to the extensiveness of the changes.

Custodians:
Army - ER
Navy - EC
Air Force - 17

Review activities:
Army - MI, AR
Air Force - 11, 19, 85, 99
Navy - OS, SH, TD
DLA - ES

User activities:
Army - SM
Navy - CG, MC, AS

Agent:
DLA - ES

Preparing activity:
Air Force - 17

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1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-M-38510/5C	MICROCIRCUITS, DIGITAL, TTL, AND-OR-INVERT GATES MONOLITHIC SILICON

3a. NAME OF SUBMITTING ORGANIZATION

b. ADDRESS *(Street, City, State, ZIP Code)*

4. TYPE OF ORGANIZATION *(Mark one)*

- ☐ VENDOR
- ☐ USER
- ☐ MANUFACTURER
- ☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
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